

Myhill Nerode Theorem

Let L be an arbitrary language over an alphabet Σ . We define a binary relation R_L on Σ^* as follows.

Def For $x, y \in \Sigma^*$, $x R_L y$
iff
 $\forall z \in \Sigma^* [xz \in L \iff yz \in L]$

Exercise: Show that R_L is an equivalence relation on Σ^* .

It is well known that an equivalence relation on set A partitions A into its equivalence classes.

Def Let R be an equivalence relation on set A . Index of R is defined as the number of distinct equivalence classes of A .

It may be noted that index of R is the same as the number of parts ~~of A~~ in the partition of A , induced by R .

In trying to design a DFA for a language L , we think about relevant information in an arbitrary word w that is ~~not~~ needed to ~~determine~~ be stored to determine membership in L of extensions of w .

For instance, if $x R_L y$ then x, y need to end-up in different states of DFA. If R_L has infinite index, then we have infinitely many words which are mutually unrelated by R_L . These all should end-up in different states of DFA, which would require DFA to have infinitely many distinct states.

Myhill Nerode theorem formalizes this intuition.

Theorem (Myhill-Nerode)

Let Σ be a finite alphabet.

$L \subseteq \Sigma^*$ is regular iff R_L is of finite index.

Proof:

$\Leftarrow$ Let R_L have equivalence classes $[x_1], \dots, [x_r]$ (where $x_i \in \Sigma^*$)

($[x_i]$ is the equivalence class of $x_i \in \Sigma^*$)

We define automaton M_L as

$(\{[x_1], \dots, [x_r]\}, \Sigma, [x_1], \delta_{M_L}, F)$

where $\delta_{M_L}([x_i], a) = [x_i a]$, $a \in \Sigma$

Exercise: show that δ_{M_L} is well defined (that is its

definition is independent of representative of the equivalence class.)

$$F = \{ [x] \mid x \in L \}$$

Exercise: In the same way as for above exercise show that set F is independent of representatives of classes chosen to define it.

To prove correctness of M_L , we ask reader to prove the following. ~~exercise~~.

Exercise: For each $x \in \Sigma^*$, show that for all $w \in \Sigma^*$

$$\hat{\delta}_{M_L}([x], w) = [xw].$$

(Proof is by an easy induction on $|w|$ ^{simultaneously} for all ~~fixed~~ x)

Now, for any $y \in \Sigma^*$,

$$y \in L(M_L) \iff \delta_{M_L}([ε], y) \in F$$

$$\iff$$

$$[y] \in F$$

$$\iff$$

$$y \in L$$

$\Rightarrow$ Let $M = (Q, \Sigma, q_0, \delta_M, F)$ be a DFA s.t.

$$L(M) = L$$

Define a relation R_M on Σ^* by

$$x R_M y \iff \delta_M(q_0, x) = \delta_M(q_0, y)$$

Exercise: Show that R_M is an equivalence relation.

We now observe that

$$x R_M y \Rightarrow x R_L y$$

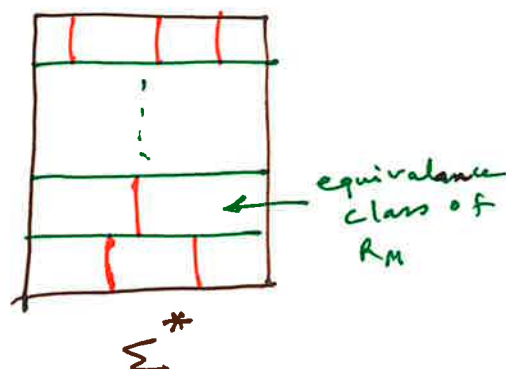
let $z \in \Sigma^*$

$$\begin{aligned}\delta_M(q_0, xz) &= \delta_M(\delta_M(q_0, x), z) \\ &= \delta_M(\delta_M(q_0, y), z) \\ &= \delta_M(q_0, yz)\end{aligned}$$

it follows, ~~$xz \in L$~~ $xz \in L \Leftrightarrow yz \in L$.

To see that R_M has at least as many equivalence classes as R_L consider the following picture.

Horizontal stripes are for equivalence classes of R_L



Each equivalence class of R_M is contained in some equivalence class of R_L

Red vertical lines within a horizontal stripe show equivalence class of R_M .

Also note that $\text{index of } R_M \leq |Q|$

$$\Rightarrow \text{index of } R_L \leq \text{index of } R_M \leq |Q|$$

(from diagram) (from def of R_M)

$\Rightarrow$ index of R_L is finite.

□

Some Examples of R_L

Example 1:

let $L = \{x \in \{a, b\}^* \mid x \text{ has equal no. of } a\text{'s and } b\text{'s}\}$

Consider $E_i = \{x \mid \#_a(x) - \#_b(x) = i\}$, where i is an integer.

It is easy to see that $\{E_i\}_{i \in \mathbb{Z}}$ partition

Σ^* . Moreover, this is the partition induced by R_L .

To see this let $y \in E_i$, then for exactly those z ,
o.t. $\#_a(z) - \#_b(z) = -i$, ~~$xz \in L$ and~~ $yz \in L$.

It follows that $y R_L x$, for any $x \in E_i$ and
 ~~$y R_L x$~~ , for any $x \notin E_i$.

Example 2:

let $L = \{x \in \{0, 1\}^* \mid \#_0(x) \bmod 5 = 0\}$

Let $E_i = \{x \in \{0, 1\}^* \mid \#_0(x) \bmod 5 = i\}$, for
 $i = 0, 1, 2, 3, 4$.

For $y \in E_i$, only those z o.t. $\#_0(z) \bmod 5 = 5 - i$
lead to $yz \in L$. It follows that E_i 's are equiv classes
of R_L . That is $x R_L y$ iff $x, y \in E_i$ for some i .

Description of R_L is not always as neat as in examples above.

Example 3:

Let $L = \{0^n 1^n \mid n \geq 1\}$

We would like to describe the partition induced by R_L .

Observe that only words of the form $\varepsilon, 0^i, 0^i 1^j$ where $j \leq i$ have a z s.t. when appended with z resulting word is in L . (In other words, all prefixes of element of L are of the form ~~as above~~ as above.)

It follows that all $w \in \{0,1\}^*$ s.t. w is not of the above form one equivalence class of R_L .

Remaining equivalence classes of R_L come from words of the form $\varepsilon, 0^i, 0^{i+j} 1^j$. We analyze these below.

$\{\varepsilon\}$ is one equivalence class.

For each $i > 0$, $\{0^i\}$ is an equivalence class of R_L .

For each $i \geq 0$, $\{0^{i+j} 1^j \mid j > 0\}$ is an equivalence class of R_L .

~~$\{0^{i+j}\}$~~

Exercise: show that $0^i \not\sim_L 0^{i+j} 1^j$, for any $j > 0$.